Project: Worked examples
Title: EN 1993-1-1 LTB and N+M interaction
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## 1. Scope and method

Lateral-torsional buckling (6.3.2), flexural buckling (6.3.1) and the interaction of axial compression with bending (6.3.3) for a uniform doubly-symmetric I-section member on fork supports, class 1 to 3: M_cr, $\bar{\lambda}_{LT}$, χ_LT and M_b,Rd; χ_y, χ_z and N_b,Rd; the Annex B (or Annex A) interaction factors and both interaction ratios, with the governing check.
Critical moment (NCCI SN003 form): $M_{cr} = C_1 \cdot \pi^2 E \cdot I_z/(kL)^2 \cdot [\sqrt{((k/k_w)^2 \cdot I_w/I_z + (kL)^2 \cdot G \cdot I_t/(\pi^2 E \cdot I_z) + (C_2 z_g)^2)} - C_2 z_g]$; $C_1 = 1.88 - 1.40\psi + 0.52\psi^2 \le 2.70$ (end moments, k = 1) unless entered LTB slenderness: $\bar{\lambda}_{LT} = \sqrt{(W_y \cdot f_y/M_{cr})}$, $W_y = W_{pl,y}$ (class 1, 2) or $W_{el,y}$ (class 3) 6.3.2.2: $\Phi_{LT} = 0.5[1 + \alpha_{LT}(\bar{\lambda}_{LT} - 0.2) + \bar{\lambda}_{LT}^2]$; $\chi_{LT} = 1/(\Phi_{LT} + \sqrt{(\Phi_{LT}^2 - \bar{\lambda}_{LT}^2)}) \le 1$ 6.3.2.3: $\Phi_{LT} = 0.5[1 + \alpha_{LT}(\bar{\lambda}_{LT} - \bar{\lambda}_{LT,0}) + \beta \cdot \bar{\lambda}_{LT}^2]$; $\chi_{LT} = 1/(\Phi_{LT} + \sqrt{(\Phi_{LT}^2 - \beta \cdot \bar{\lambda}_{LT}^2)}) \le 1, \le 1/\bar{\lambda}_{LT}^2$; optional $\chi_{LT,mod} = \chi_{LT}/f \le 1, \le 1/\bar{\lambda}_{LT}^2$ (eq. 6.58), $f = 1 - 0.5(1 - k_c)[1 - 2(\bar{\lambda}_{LT} - 0.8)^2] \le 1$ Buckling resistance in bending: $M_{b,Rd} = \chi_{LT} \cdot W_y \cdot f_y/\gamma_{M1}$ Flexural buckling: $N_{cr} = \pi^2 EI/L_{cr}^2$, $\bar{\lambda} = \sqrt{(A \cdot f_y/N_{cr})}$, $\Phi = 0.5[1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2]$, $\chi = 1/(\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \le 1$, $N_{b,Rd} = \chi \cdot A \cdot f_y/\gamma_{M1}$; the curve of each axis from Table 6.2 (section type, h/b, t_f; $f_y \le 420$ MPa) or chosen Interaction (6.3.3(4)): $N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1}) + k_{yy} \cdot M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1}) + k_{yz} \cdot M_{z,Ed}/(M_{z,Rk}/\gamma_{M1}) \le 1$; $N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1}) + k_{zy} \cdot M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1}) + k_{zz} \cdot M_{z,Ed}/(M_{z,Rk}/\gamma_{M1}) \le 1$ Factors k_ij: Annex B from the slenderness, $N_{Ed}/(\chi \cdot N_{Rk}/\gamma_{M1})$ and $C_m$ ($C_m = 0.6 + 0.4\psi \ge 0.4$ for a linear diagram, or entered); Annex A from the μ, w, $n_{pl}$, $a_{LT}$ and $C_{ij}$ expressions (class 1, 2: plastic column of Table A.1; class 3: elastic column, without the $0.6\sqrt{(w_i/w_j)}$ factors of $k_{yz}$, $k_{zy}$)

Ref: EN 1993-1-1:2005+A1:2014 6.3.1.2, 6.3.1.3 — Buckling curves; slenderness for flexural buckling
Ref: EN 1993-1-1:2005+A1:2014 6.3.2.2 — Lateral-torsional buckling curves: general case
Ref: EN 1993-1-1:2005+A1:2014 6.3.2.3 — Lateral-torsional buckling curves for rolled sections and equivalent welded sections
Ref: EN 1993-1-1:2005+A1:2014 6.3.3 — Uniform members in bending and axial compression
Ref: EN 1993-1-1:2005+A1:2014 Annex A, Annex B — Interaction factors k_ij: methods 1 and 2
Ref: NCCI SN003 Mcr expression — Elastic critical moment of a doubly-symmetric member
Ref: EN 1993-1-1:2022 8.3 — Stability of members (label only, see the edition note)

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| edition | EN 1993-1-1 edition (2005+A1:2014 until the 2022 edition is adopted; the 2022 choice changes only the label, see the sheet note) | 2005+A1:2014 | | spec: BS EN 1993-1-1:2005+A1:2014 with the UK National Annex | any |
| ltb_method | χ_LT method | 6.3.2.3 (rolled and welded sections) | | assumption: 6.3.2.3 chosen for a hot-rolled UB; 6.3.2.2 is the alternative and gives a lower M_b,Rd. | any |
| annex | Interaction factors | Annex B (method 2) | | assumption: Annex B (method 2) chosen; Annex A can be selected for comparison. | any |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| torsion | Member behaviour for Annex B | susceptible to torsional deformation (Table B.2) | | assumption: An open I-section with no torsional restraint along its length is susceptible to torsional deformation (Table B.2). | any |
| f_y | Yield strength (for the thickness) | 345 | MPa | spec: BS EN 10025-2 S355, 16 mm < t ≤ 40 mm (flange 19.6 mm) | 235 … 460 MPa |
| E | Modulus of elasticity | 210 | GPa | spec: BS EN 1993-1-1 cl. 3.2.6 | 190 … 220 GPa |
| G | Shear modulus | 81 | GPa | spec: BS EN 1993-1-1 cl. 3.2.6: G = E/(2(1 + ν)) ≈ 81 GPa | 70 … 90 GPa |
| gamma_M1 | Partial factor γ_M1 (member buckling) | 1 | | spec: UK NA to BS EN 1993-1-1, NA.2.15 | 1 … 1.25 |
| section_class | Cross-section class (class 4 needs effective properties and is outside this sheet) | 1 | | assumption: Class 1 assumed for S355 UB 457×191×98 under N + M; the engineer confirms the classification (EN 1993-1-1 Table 5.2). | any |
| section_type | Section type (selects the buckling curve of Table 6.4 / 6.5) | rolled I | | spec: BS EN 10365 UB 457×191×98, hot-rolled | any |
| h | Overall depth h | 467.2 | mm | spec: BS EN 10365 UB 457×191×98 | 50 … 2500 mm |
| b | Flange width b | 192.8 | mm | spec: BS EN 10365 UB 457×191×98 | 30 … 1000 mm |
| t_f | Flange thickness t_f (S1: t_f; selects the flexural buckling curves of Table 6.2) | 19.6 | mm | spec: BS EN 10365 UB 457×191×98 | 2 … 150 mm |
| A | Area A | 125.265 | cm^2 | reference: S1 section sheet SEC-01, UB 457×191×98 (A = 125.265 cm²) | 5 … 3000 cm^2 |
| I_y | Second moment of area, major axis I_y (S1: I_xx) | 45728 | cm^4 | reference: S1 section sheet SEC-01 (I_xx = 45727.99 cm⁴) | 50 … 5000000 cm^4 |
| I_z | Second moment of area, minor axis I_z (S1: I_yy) | 2347.02 | cm^4 | reference: S1 section sheet SEC-01 (I_yy = 2347.02 cm⁴) | 5 … 500000 cm^4 |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| I_t | Torsion constant I_t (S1: J) | 121.287 | cm^4 | reference: S1 section sheet SEC-01 (J = 121.287 cm⁴) | 0.1 … 100000 cm^4 |
| I_w | Warping constant I_w | 1.17554 | dm^6 | reference: S1 section sheet SEC-01 (I_w = 1.17554 dm⁶) | 0.00001 … 100000 dm^6 |
| W_pl_y | Plastic modulus, major axis (S1: W_pl_x) | 2232.46 | cm^3 | reference: S1 section sheet SEC-01 (W_pl,x = 2232.46 cm³) | 1 … 100000 cm^3 |
| W_el_y | Elastic modulus, major axis (S1: W_el_x) | 1957.53 | cm^3 | reference: S1 section sheet SEC-01 (W_el,x = 1957.53 cm³) | 1 … 100000 cm^3 |
| W_pl_z | Plastic modulus, minor axis (S1: W_pl_y) | 378.904 | cm^3 | reference: S1 section sheet SEC-01 (W_pl,y = 378.904 cm³) | 0.5 … 50000 cm^3 |
| W_el_z | Elastic modulus, minor axis (S1: W_el_y) | 243.467 | cm^3 | reference: S1 section sheet SEC-01 (W_el,y = 243.467 cm³) | 0.5 … 50000 cm^3 |
| L | Length between lateral restraints (fork supports) | 5 | m | drawing: FR-BM-12 rev B | 0.1 … 100 m |
| k | Effective length factor k (end rotation about the minor axis) | 1 | | assumption: Fork supports at the lateral restraints (no end restraint of the minor-axis rotation). | 0.5 … 1 |
| k_w | Warping factor k_w | 1 | | assumption: Free to warp at the restraints. | 0.5 … 1 |
| z_g | Load height z_g above the shear centre, towards the compression flange (positive destabilises) | 0 | mm | assumption: End-moment loading: no transverse load, so the load height does not enter. | -1000 … 1000 mm |
| C_2 | $C_2$ (0 for end moments; enter it for transverse loads) | 0 | | assumption: $C_2$ is 0 for end-moment loading. | 0 … 1.5 |
| ltb_curve | Buckling curve for χ_LT | auto (Table 6.4 / 6.5) | | assumption: Curve from Table 6.5 for a rolled I with h/b > 2. | any |
| use_f | Apply the f modification to χ_LT (6.3.2.3(2)) | yes | | assumption: The f modification of 6.3.2.3(2) is permitted; k_c from the ψ formula. | any |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| alpha_a0 | Imperfection factor α, curve a0 (recommended 0.13) | 0.13 | | spec: EN 1993-1-1 Table 6.1, recommended value | 0.05 … 1 |
| alpha_a | Imperfection factor α, curve a (recommended 0.21) | 0.21 | | spec: EN 1993-1-1 Table 6.1, recommended value | 0.05 … 1 |
| alpha_b | Imperfection factor α, curve b (recommended 0.34) | 0.34 | | spec: EN 1993-1-1 Table 6.1, recommended value | 0.05 … 1 |
| alpha_c | Imperfection factor α, curve c (recommended 0.49) | 0.49 | | spec: EN 1993-1-1 Table 6.1, recommended value | 0.05 … 1 |
| alpha_d | Imperfection factor α, curve d (recommended 0.76) | 0.76 | | spec: EN 1993-1-1 Table 6.1, recommended value | 0.05 … 1 |
| lambda_LT0 | $\bar{\lambda}_{LT,0}$ for 6.3.2.3 (recommended 0.4) | 0.4 | | spec: EN 1993-1-1 cl. 6.3.2.3(1), recommended value | 0.2 … 0.5 |
| beta_LT | β for 6.3.2.3 (recommended 0.75) | 0.75 | | spec: EN 1993-1-1 cl. 6.3.2.3(1), recommended value | 0.5 … 1 |
| moment_mode | Moment diagram given as | end moments | | assumption: The moments are the member end moments. | any |
| N_Ed | Design axial force, TENSION positive and compression negative (as S2 N_e1 and S3; 6.3.3 applies to compression) | -400 | kN | reference: S2 frame sheet FR-01, member 2, ULS combination (compression negative) | -100000 … 0 kN |
| M_y_A | Moment at end A about y-y, as an ordinate of the bending-moment diagram (end-moment mode): the same sign at both ends is single curvature, opposite signs double curvature. Not the end actions of a stiffness analysis, which flip sign at one end | 340 | kN*m | reference: S2 frame sheet FR-01, member 2, end A | -100000 … 100000 kN*m |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| M_y_B | Moment at end B about y-y, same convention as A: a bending-moment-diagram ordinate, the same sign as A for single curvature (end-moment mode) | 170 | kN*m | reference: S2 frame sheet FR-01, member 2, end B (same side: single curvature) | -100000 ... 100000 kN*m |
| M_z_Ed | Greatest design moment about z-z | 6 | kN*m | assumption: Small minor-axis moment from the cladding-rail eccentricity (engineer's estimate). | 0 ... 100000 kN*m |
| psi_z | ψ of the M_z diagram | 1 | | assumption: Uniform minor-axis moment (conservative). | -1 ... 1 |
| L_cr_y | Buckling length about y-y | 5 | m | drawing: FR-BM-12 rev B | 0.1 ... 100 m |
| L_cr_z | Buckling length about z-z | 5 | m | drawing: FR-BM-12 rev B | 0.1 ... 100 m |
| curve_y | Buckling curve about y-y (auto: Table 6.2 from the section type, h/b and t_f, for f_y up to 420 MPa) | auto (Table 6.2) | | spec: EN 1993-1-1 Table 6.2: rolled I, h/b > 1.2, t_f ≤ 40 mm, buckling about y-y gives curve a | any |
| curve_z | Buckling curve about z-z (auto: Table 6.2 from the section type, h/b and t_f, for f_y up to 420 MPa) | auto (Table 6.2) | | spec: EN 1993-1-1 Table 6.2: rolled I, h/b > 1.2, t_f ≤ 40 mm, buckling about z-z gives curve b | any |

## 3. Assumptions

A1. Uniform doubly-symmetric I-section member on fork supports (free to warp and to rotate about the minor axis at the ends, restrained against twist); no mono-symmetric or tapered members, no cantilevers, no continuous-restraint effects. (Scope of S7; k and k_w below 1 model partial end restraint of the minor-axis rotation and of warping.)
A2. Class 1, 2 or 3 as stated by the user; class 4 effective properties are outside this sheet. (6.2.2.5 is not implemented.)
A3. $C_1$ from the ψ formula applies to end-moment diagrams with k = 1; for transverse loads, k ≠ 1 or other diagrams the user enters $C_1$, $C_2$ (and k_c, C_m) from the source in use. With $C_1$ entered the sheet blocks until k_c (when the f modification is used) and C_my (Annex B) are entered too, unless the diagram is given as end moments: ψ is then a real, entered moment shape. (No table of $C_1$ or $C_2$ values is shipped (reference records only).)
A4. The imperfection factors, β and $\bar{\lambda}_{LT,0}$ default to the recommended values and are a user-verifiable factor set; the National Annex in use may differ. (Record kind ec3.ltb-factors.)
A5. The 2022 edition choice changes the label printed on the sheet only, by decision: the arithmetic implemented is that of EN 1993-1-1:2005+A1:2014, the differences between the editions are not implemented (they cannot be stated with certainty), and the clause numbers in the 2022 edition differ and must be checked by the user. (The sheet says so (warning `edition`).)

A6. The flexural buckling curves selected automatically are those of Table 6.2 for S 235 to S 420 (rolled I: a, b for h/b > 1.2 and t_f ≤ 40 mm, b, c above 40 mm or h/b ≤ 1.2 up to 100 mm, d, d beyond; welded I: b, c up to 40 mm, c, d above); for f_y above 420 MPa or a combination the table does not cover the curves are chosen by hand. (Table 6.2 is a small selection rule and is not reproduced beyond it.)
A7. The Annex A and Annex B expressions are implemented as published; their coefficients are to be checked against the owner's copy of the standard. (Interaction factors for diagrams other than linear are entered, not derived.)
A8. ltb_method (χ_LT method) is taken as 6.3.2.3 (rolled and welded sections). (6.3.2.3 chosen for a hot-rolled UB; 6.3.2.2 is the alternative and gives a lower M_b,Rd.)
A9. annex (Interaction factors) is taken as Annex B (method 2). (Annex B (method 2) chosen; Annex A can be selected for comparison.)
A10. torsion (Member behaviour for Annex B) is taken as susceptible to torsional deformation (Table B.2). (An open I-section with no torsional restraint along its length is susceptible to torsional deformation (Table B.2).)
A11. section_class (Cross-section class (class 4 needs effective properties and is outside this sheet)) is taken as 1. (Class 1 assumed for S355 UB 457×191×98 under N + M; the engineer confirms the classification (EN 1993-1-1 Table 5.2).)
A12. k (Effective length factor k (end rotation about the minor axis)) is taken as 1. (Fork supports at the lateral restraints (no end restraint of the minor-axis rotation).)
A13. k_w (Warping factor k_w) is taken as 1. (Free to warp at the restraints.)
A14. z_g (Load height z_g above the shear centre, towards the compression flange (positive destabilises)) is taken as 0 mm. (End-moment loading: no transverse load, so the load height does not enter.)
A15. C_2 (C₂ (0 for end moments; enter it for transverse loads)) is taken as 0. (C₂ is 0 for end-moment loading.)
A16. ltb_curve (Buckling curve for χ_LT) is taken as auto (Table 6.4 / 6.5). (Curve from Table 6.5 for a rolled I with h/b > 2.)
A17. use_f (Apply the f modification to χ_LT (6.3.2.3(2))) is taken as yes. (The f modification of 6.3.2.3(2) is permitted; k_c from the ψ formula.)
A18. moment_mode (Moment diagram given as) is taken as end moments. (The moments are the member end moments.)
A19. M_z_Ed (Greatest design moment about z-z) is taken as 6 kN*m. (Small minor-axis moment from the cladding-rail eccentricity (engineer's estimate).)
A20. psi_z (ψ of the M_z diagram) is taken as 1. (Uniform minor-axis moment (conservative).)
A21. Flexural buckling curves a (y-y) and b (z-z) from Table 6.2: rolled I-section, h/b = 2.4232, t_f = 19.6 mm. (Automatic selection; choose the curves by hand to override.)

## 4. Working

[36] M_y_Ed = max(|M_A|, |M_B|) 6.3.2
= max(|340|, |170|) kN·m
= 340 kN·m
Greatest design moment about y-y

[37] psi_y = □ = M_small/M_large (same sign: single curvature) Table B.3
= 170 / 340
= 0.5
Ratio of the end moments, taken as ordinates of the bending-moment diagram (same sign at both ends: single curvature)

[38] C_1 = 1.88 − 1.40□ + 0.52□² ≤ 2.70 NCCI SN003 / ENV 1993-1-1 Annex F
= □ = 0.5
= 1.31
$C_1$ for end moments (derived)

[39] Mcr = $C_1 \cdot \pi^2 E \cdot I_z/(kL)^2 \cdot [\sqrt{((k/k_w)^2 I_w/I_z + (kL)^2 G \cdot I_t/(\pi^2 E \cdot I_z) + (C_2 z_g)^2)} - C_2 z_g]$ NCCI SN003
= 1.31 × π² × 210000 × 23470000 / (5000)² × [√((1/1)² × 1175500000000/23470000 + (5000)² × 81
1212900 / (π² × 210000 × 23470000) + (0 × 0)²) − 0]
= 808.38 kN·m

Elastic critical moment for lateral-torsional buckling

[40] W_y = W_pl,y (class 1, 2) 6.3.2.2(1)

= plastic

= 2232.5 cm³

Section modulus used for bending about y

[41] lambda_LT = √(W_y·f_y/M_cr) 6.3.2.2(1)

= √(2232.5 cm³ × 345 MPa / 808.38 kN·m)

= 0.9761

Non-dimensional slenderness for lateral-torsional buckling

[48] alpha_LT = imperfection factor, curve c 6.3.2.3(1), Table 6.5

= Table 6.5, rolled, h/b = 2.4232

= 0.49

Imperfection factor for lateral-torsional buckling

[51] Phi_LT = 0.5[1 + □_LT(□̄_LT - □̄_LT,0) + □·□̄_LT²] 6.3.2.3(1)

= 0.5 × [1 + 0.49 × (0.9761 - 0.4) + 0.75 × 0.9761²]

= 0.99843

Φ_LT

[52] chi_LT_0 = 1/(□_LT + √(□_LT² - □·□̄_LT²)) ≤ 1, ≤ 1/□̄_LT² 6.3.2.3(1)

= 1/(0.99843 + √(0.99843² - 0.75 × 0.9761²))

= 0.6537

Lateral-torsional reduction factor

[54] k_c = 1/(1.33 - 0.33□) Table 6.6

= □ = 0.5

= 0.85837

k_c for a linear moment diagram (derived)

[55] f = 1 - 0.5(1 - k_c)[1 - 2(□̄_LT - 0.8)²] ≤ 1 6.3.2.3(2)

= 1 - 0.5 × (1 - 0.85837) × [1 - 2 × (0.9761 - 0.8)²]

= 0.93358

Modification factor f

[56] chi_LT_mod = □_LT/f ≤ 1, ≤ 1/□̄_LT² (eq. 6.58) 6.3.2.3(2)

= 0.6537 / 0.93358, 1/□̄_LT² = 1.0496

= 0.70022

Modified lateral-torsional reduction factor

[57] chi_LT = □_LT,mod 6.3.2.3

= as above

= 0.70022

Lateral-torsional reduction factor used

[58] Mb_Rd = □_LT·W_y·f_y/□_M1 6.3.2.1(3)

= 0.70022 × 2232.5 cm³ × 345 MPa / 1

= 539.3 kN·m

Design buckling resistance moment

[59] u_LTB = M_y_Ed / Mb_Rd

= 340 kN·m / 539.3 kN·m

= 0.63044

Lateral-torsional buckling utilisation

[62] alpha_y = imperfection factor, curve a 6.3.1.2, Table 6.2

= Table 6.2

= 0.21

Imperfection factor about y

[63] alpha_z = imperfection factor, curve b 6.3.1.2, Table 6.2

= Table 6.2

= 0.34

Imperfection factor about z

[64] $N_{Rk} = A \cdot f_y$ 6.3.1.1(3)
$= 125.27\ cm^2 \times 345\ MPa$
$= 4321.6\ kN$
Characteristic axial resistance
[65] $N_{cr,y} = \pi^2 E \cdot I_y / L_{cr,y}^2$ 6.3.1.2
$= \pi^2 \times 210000 \times 457280000 / 5000^2$
$= 37911\ kN$
Elastic critical force about y
[66] $N_{cr,z} = \pi^2 E \cdot I_z / L_{cr,z}^2$ 6.3.1.2
$= \pi^2 \times 210000 \times 23470000 / 5000^2$
$= 1945.8\ kN$
Elastic critical force about z
[67] $\lambda_y = \sqrt{(A \cdot f_y / N_{cr,y})}$ 6.3.1.3
$= \sqrt{(4321.6 / 37911)}$
$= 0.33763$
Non-dimensional slenderness about y
[68] $\lambda_z = \sqrt{(A \cdot f_y / N_{cr,z})}$ 6.3.1.3
$= \sqrt{(4321.6 / 1945.8)}$
$= 1.4903$
Non-dimensional slenderness about z
[69] $\chi_y = 1 / (\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \le 1$ 6.3.1.2
$= \Phi = 0.21,\ \bar{\lambda} = 0.33763$
$= 0.96853$
Reduction factor about y
[70] $\chi_z = 1 / (\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \le 1$ 6.3.1.2
$= \Phi = 0.34,\ \bar{\lambda} = 1.4903$
$= 0.34582$
Reduction factor about z
[71] $N_{b,y,Rd} = \chi_y \cdot A \cdot f_y / \gamma_{M1}$ 6.3.1.1(3)
$= 0.96853 \times 4321.6 / 1$
$= 4185.7\ kN$
Buckling resistance about y
[72] $N_{b,z,Rd} = \chi_z \cdot A \cdot f_y / \gamma_{M1}$ 6.3.1.1(3)
$= 0.34582 \times 4321.6 / 1$
$= 1494.5\ kN$
Buckling resistance about z
[73] $N_{b,Rd} = \min(N_{b,y,Rd}, N_{b,z,Rd})$
$= \min(4185.7\ kN, 1494.5\ kN)$
$= 1494.5\ kN$
Flexural buckling resistance (smaller of the two axes)
[74] $M_{y,Rk} = W_y \cdot f_y$ 6.3.3
$= 2232.5\ cm^3 \times 345\ MPa$
$= 770.2\ kN \cdot m$
Characteristic bending resistance about y
[75] $M_{z,Rk} = W_z \cdot f_y$ ($W_{pl,z}$ class 1, 2; $W_{el,z}$ class 3) 6.3.3
$= 378.9\ cm^3 \times 345\ MPa$
$= 130.72\ kN \cdot m$
Characteristic bending resistance about z
[76] $C_{my} = 0.6 + 0.4\psi_y \ge 0.4$ (or entered) Table B.3
$= \psi_y = 0.5$
$= 0.8$
Equivalent uniform moment factor, y

[77] C_mz = 0.6 + 0.4χ_z ≥ 0.4 (or entered) Table B.3
= χ_z = 1
= 1
Equivalent uniform moment factor, z
[78] C_mLT = as C_my (or entered) Table B.3
= C_my
= 0.8
Equivalent uniform moment factor for lateral-torsional buckling
[79] k_yy = Annex B: C_my(1 + (λ̄_y - 0.2)n_y) ≤ C_my(1 + 0.8n_y) (class 1, 2) Annex B Table B.2
= n_y = 0.095565
= 0.81052
Interaction factor k_yy
[80] k_yz = Annex B: 0.6k_zz (class 1, 2), k_zz (class 3) Annex B Table B.2
= as above
= 0.82482
Interaction factor k_yz
[81] k_zy = Annex B: Table B.1 0.6k_yy / 0.8k_yy; Table B.2 from λ̄_z, n_z and C_mLT Annex B Table B.1
= n_z = 0.26764
= 0.95134
Interaction factor k_zy
[82] k_zz = Annex B: C_mz(1 + (2λ̄_z - 0.6)n_z) ≤ C_mz(1 + 1.4n_z) (class 1, 2) Annex B Table B.2
= as above
= 1.3747
Interaction factor k_zz
[83] u_N_y = N_Ed/(χ_y·N_Rk/γ_M1) 6.3.3(4)
= 400 kN / (0.96853 × 4321.6 kN / 1)
= 0.095565
Axial term about y
[84] u_N_z = N_Ed/(χ_z·N_Rk/γ_M1) 6.3.3(4)
= 400 kN / (0.34582 × 4321.6 kN / 1)
= 0.26764
Axial term about z
[85] u_M_y = M_y,Ed/(χ_LT·M_y,Rk/γ_M1) 6.3.3(4)
= 340 kN·m / (0.70022 × 770.2 kN·m / 1)
= 0.63044
Major-axis bending term
[86] u_M_z = M_z,Ed/(M_z,Rk/γ_M1) 6.3.3(4)
= 6 kN·m / (130.72 kN·m / 1)
= 0.045899
Minor-axis bending term
[87] u_61 = N_Ed/(χ_y N_Rk/γ_M1) + k_yy·M_y,Ed/(χ_LT M_y,Rk/γ_M1) + k_yz·M_z,Ed/(M_z,Rk/γ_M1) 6.3.3(4)
= 0.095565 + 0.81052 × 0.63044 + 0.82482 × 0.045899
= 0.64441
Interaction, equation 6.61
[88] u_62 = N_Ed/(χ_z N_Rk/γ_M1) + k_zy·M_y,Ed/(χ_LT M_y,Rk/γ_M1) + k_zz·M_z,Ed/(M_z,Rk/γ_M1) 6.3.3(4)
= 0.26764 + 0.95134 × 0.63044 + 1.3747 × 0.045899
= 0.9305
Interaction, equation 6.62
[89] u_interaction = max(u_61, u_62) 6.3.3(4)
= max(0.64441, 0.9305)
= 0.9305
Greatest interaction ratio

[90] u_buckling = N_Ed/N_b,Rd 6.3.1.1
= 400 kN / 1494.5 kN
= 0.26764
Flexural buckling utilisation
[91] u_member = max(max(u_interaction, u_LTB), u_buckling)
= max(max(0.9305, 0.63044), 0.26764)
= 0.9305
Governing utilisation of the member

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| edition_used | "EN 1993-1-1:2005+A1:2014" | EN 1993-1-1 edition |
| psi_y | 0.5 | ψ of the M_y diagram |
| M_y_Ed | 340 kN·m | Greatest design moment about y-y |
| C_1 | 1.31 | $C_1$ |
| Mcr | 808.38 kN·m | M_cr |
| lambda_LT | 0.9761 | λ̄_LT |
| chi_LT | 0.70022 | χ_LT |
| Mb_Rd | 539.3 kN·m | M_b,Rd |
| N_Rk | 4321.6 kN | N_Rk |
| N_cr_y | 37911 kN | N_cr,y |
| N_cr_z | 1945.8 kN | N_cr,z |
| lambda_y | 0.33763 | λ̄_y |
| lambda_z | 1.4903 | λ̄_z |
| chi_y | 0.96853 | χ_y |
| chi_z | 0.34582 | χ_z |
| N_b_y_Rd | 4185.7 kN | N_b,y,Rd |
| N_b_z_Rd | 1494.5 kN | N_b,z,Rd |
| N_b_Rd | 1494.5 kN | N_b,Rd |
| k_yy | 0.81052 | k_yy |
| k_yz | 0.82482 | k_yz |
| k_zy | 0.95134 | k_zy |
| k_zz | 1.3747 | k_zz |
| u_LTB | 0.63044 | LTB utilisation |
| u_buckling | 0.26764 | Buckling utilisation |
| u_61 | 0.64441 | Equation 6.61 |
| u_62 | 0.9305 | Equation 6.62 |
| u_interaction | 0.9305 | Interaction utilisation |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| u_member | 0.9305 | Member utilisation |
| alpha_LT | 0.49 | α_LT |
| alpha_y | 0.21 | α_y |
| alpha_z | 0.34 | α_z |
| Phi_LT | 0.99843 | Φ_LT |
| chi_LT_0 | 0.6537 | χ_LT before the f modification |
| k_c | 0.85837 | k_c |
| f | 0.93358 | f |
| W_y | 2232.5 cm³ | W_y |
| M_y_Rk | 770.2 kN·m | M_y,Rk |
| M_z_Rk | 130.72 kN·m | M_z,Rk |
| C_my | 0.8 | C_my |
| C_mz | 1 | C_mz |
| C_mLT | 0.8 | C_mLT |
| u_N_y | 0.095565 | Axial term, y |
| u_N_z | 0.26764 | Axial term, z |
| u_M_y | 0.63044 | Bending term, y |
| u_M_z | 0.045899 | Bending term, z |
| governing | "interaction 6.62 (0.931)" | Governing check |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Interaction, equation 6.62 | u_62 = 0.9305 | u_62 ≤ 1 | 0.931 | 6.3.3(4) |
| PASS | Interaction, equation 6.61 | u_61 = 0.64441 | u_61 ≤ 1 | 0.644 | 6.3.3(4) |
| PASS | Lateral-torsional buckling M_y,Ed ≤ M_b,Rd | u_LTB = 0.63044 | u_LTB ≤ 1 | 0.63 | 6.3.2.1 |
| PASS | Flexural buckling N_Ed ≤ N_b,Rd | u_buckling = 0.26764 | u_buckling ≤ 1 | 0.268 | 6.3.1.1 |

## 7. Applicability

WARNING — record: Tabulated values not taken from a reference record: E, G, f_y (Material properties); gamma_M1 (EN 1993-1-1 National Annex factors); alpha_a0, alpha_a, alpha_b, alpha_c, alpha_d, lambda_LT0, beta_LT (Imperfection factors and 6.3.2.3 parameters); C_2 ($C_1$, $C_2$ and k_c for the moment diagram). Enter each as one so its source, edition and licence travel with it.

## 8. Verification

Template struct.ltb-interaction@1.1.0 Content hash aba6cc96a341bb38 Engine calc-engine/1.0.0
Known-answer cases passed 8/8 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0